

Product Requirements Document (PRD)

Threads Affiliate Automation Platform

Dual-Platform: Shopee Affiliate + TikTok Shop Affiliate
Auto-Posting Organik ke Threads (Meta)
Manual Link Input + AI Content + Auto Publish

Versi:	2.0.0 (Audit-Fixed)
Tanggal:	06 May 2026
Author:	enowX Labs
Status:	Final
Publishing Platform:	Threads (Meta)
Affiliate Sources:	Shopee + TikTok Shop
Link Input:	Manual (user-generated)
AI Engine:	Claude API (content gen)
Job Queue:	Asynq (Go-native)

1. Executive Summary

Platform web app berbasis AI yang mengotomasi affiliate marketing melalui Threads (Meta). Mendukung dual-platform affiliate: Shopee Affiliate dan TikTok Shop Affiliate. User menginput link affiliate secara manual, lalu AI handle content generation dan auto-posting yang terlihat organik.

v2.0 Changes (Post-Audit):

- Fixed: Job queue dari Bull (Node.js) ke Async (Go-native)
- Fixed: Product info extraction = best-effort + manual override (bukan dependency)
- Fixed: Self-learning optimize CLICKS & ENGAGEMENT (bukan conversions)
- Added: Legal & Compliance section (Threads API ToS)
- Added: Security & Authentication detail
- Added: Account safety circuit breaker
- Added: Testing strategy
- Fixed: Post limits konsisten (max 25/day per account)
- Fixed: Revenue projection lebih konservatif
- Fixed: AI cost estimation realistis

1.1 How It Works

1. User generate affiliate links di Shopee/TikTok dashboard (5 menit/hari)
2. Paste links ke platform (bulk supported, auto-detect platform)
3. AI generate konten organik + sisipkan link + schedule posting
4. Auto-post ke Threads via official API (max 25 posts/hari)
5. Track clicks via custom short URL (real-time)
6. AI optimize berdasarkan clicks & engagement (weekly)

User effort: ~5 menit/hari (paste links)

Automation: 95% sisanya

Optimization target: CLICKS & ENGAGEMENT (bukan conversions)

1.2 Dual-Platform Affiliate

Aspek	Shopee Affiliate	TikTok Shop Affiliate
Dashboard	affiliate.shopee.co.id	affiliate.tiktok.com
Link format	s.shopee.co.id/xxx	vt.tiktok.com/xxx
Komisi rata-rata	5-15%	5-20%
Input method	Manual paste/CSV	Manual paste/CSV
AI content tone	"Checkout di Shopee..."	"Viral di TikTok..."

1.3 Target User

- Affiliate marketer Shopee/TikTok yang ingin scale via Threads

- Content creator text-based yang mau monetisasi tanpa video
- Side-hustler yang ingin passive income (5 menit/hari)

2. Legal & Compliance

CRITICAL SECTION: Platform ini beroperasi di grey area. Compliance dengan Meta ToS adalah existential risk.

2.1 Threads API Terms of Service

Meta Threads API requirements yang HARUS dipatuhi:

- Threads API Permissions needed: threads_basic, threads_content_publish, threads_manage_insights, threads_read_replies
- Content must be "authentic" - Meta prohibits inauthentic behavior
- Automated posting is ALLOWED via API (that is the purpose of the API)
- BUT: content must not be misleading, spam, or violate community guidelines
- Affiliate disclosure: FTC requires disclosure, Meta recommends transparency

2.2 Compliance Strategy

Bagaimana platform ini comply dengan Meta ToS:

1. Human-in-the-loop DEFAULT: user harus approve konten sebelum posting (auto-mode = opt-in setelah 7 hari)
2. Content authenticity: AI generate berdasarkan REAL user preferences & niche, bukan generic spam
3. Rate limiting conservative: max 25 posts/day (jauh di bawah limit 250)
4. Affiliate disclosure: setiap post dengan link include subtle disclosure ("affiliate link")
5. No fake engagement: tidak auto-like/follow/comment akun lain
6. User responsibility: ToS platform menyatakan user bertanggung jawab atas akun mereka

2.3

Sistem proteksi untuk mencegah ban massal:

Trigger	Action	Recovery
1 account flagged/restricted	Pause that account 48h	Reduce frequency 50% on resume
2+ accounts flagged in 24h	Pause ALL accounts 24h	Alert user, review content
API app rate limited by Meta	Emergency stop all publishing	Contact Meta, review compliance
Post rejected by Threads	Skip + log + regenerate safer	AI learns to avoid trigger
Engagement drops >50%	Reduce posting frequency	Analyze if shadowbanned

2.4 User Terms of Service

User harus agree sebelum menggunakan platform:

- User bertanggung jawab penuh atas akun Threads mereka

- User menjamin link affiliate yang diinput adalah milik mereka
- Platform tidak bertanggung jawab atas account suspension
- User harus comply dengan Shopee/TikTok affiliate program ToS
- Platform berhak pause/stop service jika detect abuse

2.5

- Click tracking: IP di-hash (tidak simpan raw IP), retain 90 hari
- OAuth tokens: encrypted at rest (AES-256)
- User data: email + preferences only, no sensitive personal data
- Data deletion: user bisa request full data deletion
- No data sharing with third parties

3. Core Features

3.1 Affiliate Link Management (Manual Input)

Fitur	Deskripsi	Prioritas
Single Link Input	Paste 1 link, auto-detect platform	P0
Bulk Paste	Multiple links (newline separated)	P0
CSV Upload	Upload CSV (nama, link, kategori, harga)	P0
Auto-detect Platform	Detect Shopee/TikTok dari URL pattern	P0
Custom Short URL	Generate go.domain.com/slug per link	P0
Manual Product Info	User input nama + kategori + harga	P0
Auto Product Info (best-effort)	Try fetch OG tags, fallback to manual	P1
Link Health Check	Periodic HTTP check, alert if expired	P1
Browser Extension	Click button on product page > save	P2

IMPORTANT: Product info extraction dari URL adalah BEST-EFFORT.
Primary method = user input manual (nama, kategori, harga).
Auto-fetch OG tags = bonus jika berhasil, bukan dependency.

3.2 AI Content Generator

6 personas, Threads-optimized, BLEND framework:

Persona	Tone	Best For
Honest Friend	Casual, jujur, relatable	Daily use reviews
Hot Take	Edgy, opinionated, bold	Comparison, debate
Problem Solver	Helpful, empathetic	Before/after stories
Curious Explorer	Excited, genuine	New discoveries
Lifestyle Sharer	Aesthetic, aspirational	Routine integration
Comparison Nerd	Analytical, fair	Rankings, side-by-side

3.3 Smart Auto-Posting

Publishing via official Threads API dengan safety measures:

Parameter	Value	Rationale
Max posts/day/account	25	Conservative (API limit 250)
Max posts/hour	4	Human-like pacing

Min interval between posts	45 minutes	Avoid burst pattern
Random delay per post	0-120 seconds	Anti-pattern detection
Max affiliate posts in row	2	Then 1-2 value posts
Content mix ratio	60% value / 40% affil	Organic feel
New account ramp-up	5/day, +3/day per we	Gradual scaling
Weekend reduction	30% fewer posts	Natural human pattern

3.4 Self-Learning AI (Click & Engagement Optimized)

IMPORTANT: AI optimizes for CLICKS and ENGAGEMENT, NOT conversions.
Karena tidak ada API untuk track conversions dari Shopee/TikTok,
AI fokus pada metrik yang bisa di-track secara real-time.

Metrik yang di-optimize oleh AI:

Metric	Source	Trackable?	AI Uses For
Post likes	Threads API	Yes (real-time)	Content quality signal
Post replies	Threads API	Yes (real-time)	Engagement trigger quality
Post reposts	Threads API	Yes (real-time)	Viral potential signal
Link clicks	Internal shortener	Yes (real-time)	PRIMARY optimization target
CTR (clicks/views)	Calculated	Yes	Content + CTA effectiveness
Follower growth	Threads API	Yes (daily)	Account health signal
Conversions	User manual input	NO (optional)	Bonus data if available
Revenue	User manual input	NO (optional)	Bonus data if available

Self-learning loop (weekly):

1. Collect: aggregate clicks + engagement per post dari 7 hari terakhir
2. Segment: group by persona, format, time, platform, category
3. Rank: top 20% vs bottom 20% posts - apa bedanya?
4. Learn: extract winning patterns (hook type, length, CTA placement)
5. Adjust: update content generation weights untuk minggu depan
6. Test: plan 1 A/B test per minggu (1 variable only)
7. Report: weekly insight ke user dashboard

3.5

Revenue tracking adalah OPTIONAL feature. User bisa input data dari affiliate dashboard:

- Weekly batch input: total revenue per minggu (simple)
- Per-product input: match revenue ke produk tertentu (detailed)
- Dashboard scraping: REMOVED from scope (terlalu fragile & risky)

Revenue data = nice-to-have untuk user insight.
AI TIDAK depend on revenue data untuk self-learning.
AI optimize clicks & engagement yang bisa di-track real-time.

4. Technical Architecture

4.1 System Stack (Fixed)

Component	Technology	Purpose
Frontend	React 18 + Vite + TailwindCSS	Dashboard & link management
Backend API	Go (Fiber framework)	Core logic, API gateway
AI Service	Python + FastAPI + Claude API	Content generation
Job Queue	Asynq (Go-native, Redis-backed)	Task scheduling & processing
Database	PostgreSQL 16	Primary data store
Cache	Redis 7	Rate limiting, click counters
URL Shortener	Go service + Redis + Cloudflare CDN	Fast redirect + tracking
Auth	JWT + Meta OAuth 2.0	User auth + Threads connection
Deploy	Docker Compose + Nginx	Production deployment
Monitoring	Prometheus + Grafana	Metrics & alerting

FIXED: Job queue changed from Bull (Node.js) to Asynq (Go-native).
Asynq uses Redis as backend, same as Bull, but native Go integration.
No need for separate Node.js worker service.

4.2 Authentication & Security

Detail auth & security implementation:

Aspect	Implementation	Notes
User auth	JWT (access + refresh tokens)	Access: 15min, Refresh: 7 days
Threads OAuth	Meta OAuth 2.0 flow	Scopes: threads_basic, content_publish, insights
Token storage	AES-256 encrypted in DB	Encryption key in env var
API rate limiting	Redis-based per user per plan	Free:100/h, Starter:500/h, Pro:2000/h
CORS	Whitelist frontend domain only	No wildcard
Input validation	Go validator + sanitization	Prevent XSS/injection
HTTPS	Enforced (redirect HTTP>HTTPS)	Let Encrypt via Nginx

4.3 Database Schema

Table	Key Fields	Purpose
users	id, email, password_hash, plan, created_at	User accounts

threads_accounts	id, user_id, threads_user_id, access_token(encrypted)	Threads connections
products	id, user_id, name, price, category, platform, image	Product catalog
affiliate_links	id, product_id, original_url, short_slug, platform, status	Links (manual input)
posts	id, account_id, link_id, content, link_placement, scheduled_at	Post queue & history
post_analytics	id, post_id, likes, replies, reposts, views, fetched_at	Engagement data
click_logs	id, link_id, hashed_ip, user_agent, referrer, created_at	Click tracking
content_templates	id, persona, format, hook_type, performance_score	AI learning data
circuit_breaker	id, account_id, event_type, triggered_at, resolved_at	Safety tracking

4.4 API Endpoints

Endpoint	Method	Auth	Purpose
/api/auth/register	POST	None	User registration
/api/auth/login	POST	None	Login (returns JWT)
/api/auth/threads/connect	GET	JWT	Start Meta OAuth flow
/api/links	POST	JWT	Add affiliate link(s)
/api/links/bulk	POST	JWT	Bulk paste links
/api/links/csv	POST	JWT	Upload CSV
/api/links	GET	JWT	List links + stats
/api/links/{id}	DELETE	JWT	Remove link
/api/posts	GET	JWT	List posts (all statuses)
/api/posts/{id}/approve	POST	JWT	Approve for publishing
/api/posts/generate	POST	JWT	Trigger AI generation
/api/analytics/overview	GET	JWT	Dashboard stats
/api/analytics/clicks	GET	JWT	Click analytics
/go/{slug}	GET	None	Short URL redirect (public)

4.5 Plan Limits (Consistent)

Plan	Harga	Links	Posts/hari	Accounts	Features
Free (7-day trial)	Rp 0	10	5	1	Basic AI, manual approve only
Starter	Rp 149K/bln	50	15	1	Full AI, auto-mode, basic analytics
Pro	Rp 399K/bln	200	25	3	Priority AI, full analytics, A/B test
Agency	Rp 999K/bln	Unlimited	25/acct	10	API access, white-label

FIXED: Post limits konsisten di semua dokumen.
Max 25 posts/day/account untuk SEMUA plan (safety limit).
Plan difference = jumlah links, accounts, dan features.
Free tier = 7-day trial (bukan permanent free).

4.6 AI Cost Estimation (Realistic)

Scale	Users	Posts/day	Tokens/day	AI Cost/month
MVP (month 1-3)	20 paid	300	~300K tokens	Rp 500K-1jt (\$30-60)
Growth (month 4-6)	100 paid	1,500	~1.5M tokens	Rp 3-5jt (\$180-300)

Scale (month 7-9)	400 paid	6,000	~6M tokens	Rp 12-20jt (\$720-1200)
Mature (month 10-12)	1000 paid	15,000	~15M tokens	Rp 30-50jt (\$1800-3000)

Mitigation: response caching (same product = cached content), batch generation, prompt optimization.

5. Product Info Extraction Strategy

FIXED: Product info extraction = BEST-EFFORT, bukan hard dependency.
User SELALU bisa input manual. Auto-fetch = bonus.

5.1 Extraction Priority

Priority	Method	Reliability	When Used
1 (Primary)	User manual input	Sangat tinggi	Always available
2 (Auto)	HTTP HEAD + OG meta tags	Medium (50-70%)	Try on link paste
3 (Fallback)	AI infer dari URL slug/path	Low (30-50%)	If OG fails
4 (Skip)	Leave empty, AI generates generic	N/A	If all fails

5.2 What Happens When Auto-Fetch Fails

Jika system tidak bisa extract product info dari URL:

1. Link tetap disimpan dengan status "needs_info"
2. User mendapat notif: "3 links perlu info tambahan"
3. User bisa input manual: nama produk + kategori + harga
4. ATAU user bisa skip > AI generate generic content berdasarkan URL saja
5. AI tetap bisa generate konten meskipun hanya punya link (tanpa detail produk)

Contoh AI tanpa product info:
"Baru nemu ini di Shopee, worth checking out sih. Link: [short_url]"

Contoh AI dengan product info:
"Serum Vitamin C 89rb ini rating 4.9 dari 15rb review. Gue udah coba 2 minggu..."

5.3 OG Tag Extraction (Best-Effort)

Technical approach untuk auto-fetch:

- HTTP GET ke affiliate URL (follow redirects, max 3 hops)
- Parse HTML response untuk: og:title, og:description, og:image
- Timeout: 5 seconds (jika lebih, skip)
- If response requires JavaScript rendering: SKIP (no headless browser in MVP)
- Success rate estimate: Shopee ~60%, TikTok ~40% (banyak SPA)
- Phase 2: optional Playwright service untuk higher success rate

6. User Interface

6.1 Key Pages

Page	Primary Action	Key Components
Dashboard	Overview at a glance	Clicks today, posts pipeline, top links
Link Manager	Add & manage links	Bulk paste, CSV, health status, stats
Content Studio	Review AI posts	Preview, edit, approve/reject, bulk
Schedule	View posting calendar	Timeline, reschedule, pause/resume
Analytics	Track clicks & engagement	CTR charts, per-link stats, trends
Accounts	Manage Threads accounts	Connect, persona, circuit breaker status
Settings	Configuration	Preferences, billing, API keys, data export

6.2 Onboarding Flow

1. Register (email + password)
2. Connect Threads account (Meta OAuth)
3. Select niche & persona (multi-select)
4. Set mode: Manual Review (default) atau Auto Mode
5. Input first 5 affiliate links (required to start)
6. AI generate 5 sample posts > user review
7. Approve > first posts scheduled > pipeline active
8. After 7 days: option to switch to Auto Mode

6.3

1. Open Shopee/TikTok Affiliate dashboard (external)
2. Generate links for 5-10 products
3. Paste links ke platform (bulk paste)
4. Optional: review pending posts (if manual mode)
5. Glance at dashboard: clicks today, top performers

7. Development Roadmap

7.1 Phase 0: Validation (Week 1)

BEFORE coding, validate critical assumptions:

- Test Threads API: create app, get token, publish test post
- Confirm: automated posting via API is allowed by Meta
- Test: Shopee/TikTok short URL resolution (can we get OG tags?)
- Setup: development environment (Go, React, PostgreSQL, Redis)
- Decision: confirm tech stack works end-to-end

DO NOT proceed to Phase 1 until Threads API publishing is confirmed working.

7.2 Phase 1: MVP (Week 2-4)

Goal: User input links > AI generate > auto-post ke Threads

- Auth: register, login (JWT), Threads OAuth connection
- Link Manager: single paste + bulk paste + platform detection
- URL shortener: generate short URL + click tracking
- AI content generation: 2 personas, single post format
- Scheduler + publisher: Asynq jobs, Threads API publish
- Basic dashboard: link list, post list, click count
- Manual approve mode (default)
- Deliverable: 1 account posting 10-15x/day from manual links

7.3

- CSV upload + auto product info (best-effort)
- 6 personas + thread format + reply drop
- Auto-reply engine (detect "beli dimana?" > reply with link)
- Analytics dashboard (clicks, CTR, engagement per link/post)
- A/B testing (2 variants per product)
- Link health monitoring
- Auto mode option (after 7 days)

7.4

- Multi-account management (up to 10)
- Self-learning AI (weekly optimization loop)
- Circuit breaker system
- Cross-post to Instagram (optional)
- Advanced scheduling (AI-optimized timing)
- Weekly insight reports

7.5

- Multi-tenant architecture
- Subscription billing (Midtrans)
- Landing page + onboarding docs
- Beta launch (50 users)
- Iterate based on feedback

8. Business Model (Conservative)

8.1 Revenue Projection (Realistic)

Timeline	Users (Trial/Paid)	Paid Conv.	MRR	Total
Bulan 1-3	200 / 10	5%	Rp 2.5jt	Rp 2.5jt
Bulan 4-6	800 / 40	5%	Rp 10jt	Rp 10jt
Bulan 7-9	2000 / 150	7.5%	Rp 40jt	Rp 40jt
Bulan 10-12	5000 / 400	8%	Rp 110jt	Rp 110jt

*FIXED: Paid conversion rate 5-8% (industry avg 2-5% untuk freemium).
Kita sedikit di atas karena free tier = 7-day trial (bukan permanent free).
Projection lebih konservatif dari v1.1.*

8.2 Cost Structure

Item	Month 1-3	Month 4-6	Month 10-12
Server (VPS)	Rp 500K	Rp 1.5jt	Rp 5jt
AI API (Claude)	Rp 1jt	Rp 5jt	Rp 40jt
Domain + CDN	Rp 100K	Rp 200K	Rp 500K
Monitoring	Rp 0 (self-hosted)	Rp 300K	Rp 1jt
TOTAL	Rp 1.6jt	Rp 7jt	Rp 46.5jt
Break-even at	~8 paid users		

8.3 Pricing Strategy

- Free tier = 7-day trial (NOT permanent free) > forces conversion decision
- Starter (Rp 149K): sweet spot untuk individual affiliate marketer
- Pro (Rp 399K): untuk serious marketer yang mau scale
- Agency (Rp 999K): untuk agency yang manage multiple clients
- Annual discount: 2 bulan gratis (Rp 1.49jt/tahun untuk Starter)

9. Testing Strategy

9.1 Testing Layers

Layer	Tool	Coverage Target	What to Test
Unit Tests	Go: testing pkg	80%+ core logic	Link parsing, scheduling, URL shortener
Unit Tests (AI)	Python: pytest	70%+ AI service	Prompt building, response parsing
Integration Tests	Go: testcontainers	All API endpoints	DB operations, Redis, auth flow
E2E Tests	Playwright	Critical user flows	Onboarding, link input, post approval
API Contract	OpenAPI spec	100% endpoints	Request/response validation
Load Tests	k6	URL shortener	Redirect latency under load

9.2 Critical Test Scenarios

- Threads API publish: mock API for unit tests, real API for staging
- URL shortener redirect: must be < 50ms p99 latency
- Concurrent click tracking: 1000 clicks/second without data loss
- Link platform detection: test all known URL patterns
- Circuit breaker: simulate account flag > verify pause behavior
- AI content: verify no hard-sell language in generated content
- Rate limiting: verify plan limits enforced correctly

9.3

- Separate Threads API app (test mode) for staging
- Separate database (seeded with test data)
- AI service with mocked responses (save API cost)
- Full pipeline test: link input > generate > publish > track click

10. Risks & Mitigations (Updated)

Risk	Impact	Prob.	Mitigation
Meta bans automated af	Critical	Low	Use official API, comply ToS, human-in-loop default
Account suspension	High	Med	Circuit breaker, gradual ramp, conservative limits
Threads API deprecation	Critical	Low	Abstraction layer, pivot to IG/X if needed
Low CTR	Medium	Med	Self-learning AI, A/B testing, persona optimization
AI cost overrun	Medium	Med	Response caching, batch gen, prompt optimization
User churn	High	Med	7-day trial, quick time-to-value, weekly reports
Competitor	Medium	High	First-mover, AI moat, user data lock-in

11. Conclusion

Platform ini menggabungkan kemudahan (manual link input, 5 menit/hari) dengan kekuatan AI (auto-generate konten organik, auto-post, self-learning). Dengan approach yang compliance-first dan conservative rate limiting, platform ini sustainable untuk jangka panjang.

Key Decisions (Post-Audit)

Decision	Rationale
Manual link input (no scraping)	Reliable, no ToS violation, user controls quality
Human-in-the-loop default	Compliance with Meta ToS, user trust, quality control
Optimize clicks not conversions	Clicks trackable real-time, conversions need manual input
Asynq (Go-native) for queue	No Node.js dependency, native Go integration
Product info = best-effort	Shopee/TikTok SPA makes scraping unreliable
7-day trial (not free tier)	Forces conversion decision, better unit economics
Max 25 posts/day/account	Conservative safety margin, avoid Meta flags
Circuit breaker system	Protect all users if one account gets flagged
No dashboard scraping	Too fragile, violates affiliate ToS, maintenance burden

IMPLEMENTATION READY: Proceed to Phase 0 (validation) then Phase 1 (MVP).
Estimated time to first working prototype: 4 weeks.
Estimated time to beta launch: 14 weeks.